

Wilfred Morales

Backend Developer — PHP/Laravel · PostgreSQL · Credit Fintech

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PROFILE

Backend developer at a payroll-deduction lending fintech — loans repaid through automatic salary deductions, a regulated sector in Colombia. I work across the credit lifecycle, from origination to identity validation and scoring, and on the platform underneath it: containerisation, layered architecture and access control.

EXPERIENCE

Backend developer *GAF Technology Solutions* · Fixed-term project contract · Remote, Colombia · Oct 2025 – Jul 2026

Payroll-deduction lending fintech. Technology team of 4: 2 developers, a tech lead, and an infrastructure and security lead.

Applications are originated by CK Comercializadora, a company in the same group.

Credit rules engine CK Comercializadora

- Modelled the data and translated business policy into a rules schema in PostgreSQL: eligibility, garnishments, term limits, and employment and financial requirements per payer. The business team now adjusts parameters without a deployment.
- Implemented the full evaluation logic: borrowing-capacity calculation under Law 1527 (payroll-deduction lending) and Law 50 for active employees and pensioners, decision criteria, special rules, and application re-evaluation.
- Built the public application portal and its front-end and back-end validation, with per-advisor traceability and reCAPTCHA Enterprise. A colleague built the submission API; I restructured its request body as the model changed.

Laravel · Vue.js · Inertia.js · PostgreSQL

Electronic signature with identity validation

- Built the electronic signature module from scratch, replacing a disused legacy one: the document is 24 pages that used to be filled in by hand and are now generated automatically from templates, with a different template per product.
- Designed the flow as a state machine over the asynchronous responses from the provider: an advisor approves a draft, customer data is validated against TransUnion, and identity is verified by OTP or KBA.
- Closed the whole procedure in about 20 minutes, with the signed document returned with its hash, full transaction traceability and a copy kept in S3.

Laravel · Vue.js · Python · TransUnion · AWS S3

Monthly credit scoring process

- Took a risk model living in a script owned by the risk team to production: a containerised monthly cron job that reads from PostgreSQL, runs the model and persists results. Scoring was not automated before this.
- Processed over 320,000 distinct people per run, loading in batches of 5,000, with an idempotent and resumable process and invalid-row handling that does not abort the load.

Python · PostgreSQL · Docker · Laravel

Access control and permissions

- Built the platform permission tree across 17 modules, with permissions down to the option and sub-process level, locked down every route, and migrated existing users and roles to the new schema without interrupting operations.

Spatie Permission · Laravel · PostgreSQL

Migration to a layered architecture

- Took part in the first stage of migrating the platform — six years of code — to a layered architecture, verifying every change module by module before moving on and removing raw SQL queries from controllers entirely. Three months of work, reviewed by the tech lead at each step; there was no test suite at the time. The second stage ran after I left.

Layered architecture · Laravel · PostgreSQL

Infrastructure and deployment

- Containerised the application with Docker, which had been running natively on the machine, and deployed it on a new GCP instance with nginx as a reverse proxy between the containers and the host.
- Ran the migration in parallel: brought up and validated the new instance while the old one was still in production, configured the domain DNS and issued the certificates with Certbot until the application ran entirely over HTTPS. No service interruption.

Docker · Linux / Nginx · GCP · Certbot

PROJECTS

Portfolio assistant

Personal project · Ongoing

- Built a conversational assistant that answers questions about my background and the decisions behind each project, with an explicit notice that an agent is answering and can be wrong.
- Rebuilt the orchestration in Python on deployment, because the Vercel free plan does not support n8n; dropped litellm after measuring it (133 MB of 199, against a 250 MB per-function ceiling) and talk to the model over plain REST with httpx.

Python · FastAPI · DeepSeek · Vercel

Video clipping CLI

Personal project · Jul 2026 – ongoing

- Built a TypeScript command-line tool that splits long video into vertical clips: Whisper transcription, script generation by a model whose output is schema-validated, speech synthesis, and subtitle rendering with Remotion.
- Persisted every step to disk and made the process resumable, so already-paid model calls are never repeated.

TypeScript · Node.js · Remotion · ffmpeg

Portfolio

Personal project · 2026

- Built this site with React and TypeScript, with variable type and animations written by hand, reinterpreting an existing design system and crediting it in the footer.

React · TypeScript · Vite

SKILLS

Backend PHP / Laravel · PostgreSQL / SQL · MySQL · Python · Node.js · Testing (PHPUnit)

Architecture Layered architecture (domain, application, persistence)

Access control Roles and permissions (Spatie) · reCAPTCHA Enterprise

Automation & AI DeepSeek API · OpenAI API · n8n · API and webhook integration

Frontend Vue.js · Inertia.js · Blade · Tailwind CSS · React · TypeScript

Infrastructure Docker · Linux / Nginx · GCP · AWS S3 · Certbot

EDUCATION

Computer Engineering, Universidad Nacional Experimental del Táchira (UNET). Currently in 7th semester.

LANGUAGES AND CERTIFICATIONS

Spanish (native) · English B2, C2 reading (EF SET) · Google IT Automation with Python — Coursera (in progress)